

The Bottleneck Was Always Human

AI-Assisted Catalog Modeling in Multi-Vendor Networks

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THE NETWORK AUTOMATION CONFERENCE

Codaxy

- 15+ years building telecom software solutions
- Creator of **CxOrchestrator**, TM Forum Gold Certified platform
- Supporting Tier 1 & Tier 2 operators globally

Everything starts with the catalog.

Everything starts with the catalog.
But the catalog is not the network.

Network abstraction

project start

delivery

network abstraction

- The network engineer new to TMF, learning the framework as they model
- No guiding principles yet, the catalog takes shape however makes sense to them

Network abstraction

project start

delivery

network abstraction

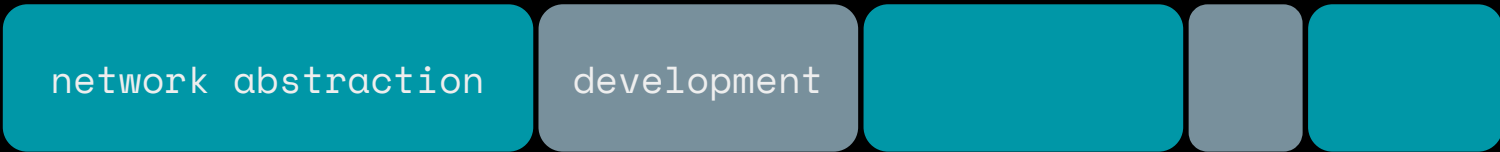
development

- The catalog made sense to the expert, but the specifications were shaped by what looks good

Network abstraction

project start

delivery



network abstraction

development

- The catalog doesn't work well with workflows
- Resource graphs too difficult to reason about at scale

Network abstraction

project start

delivery



- New use case

Network abstraction

project start

delivery



- The catalog *still* doesn't work well with workflows

So we built a tool

Why not just automate it?

- Rules about what makes a catalog good are **deeply context-specific**
 - We would also need to encode the domain knowledge, expertise and the wider business context

Because no agent
has spent years in
the field.

What we actually did

- Went back through previous projects to understand:
 - the business context
 - the design decisions
 - the revisions
 - how the catalog held up

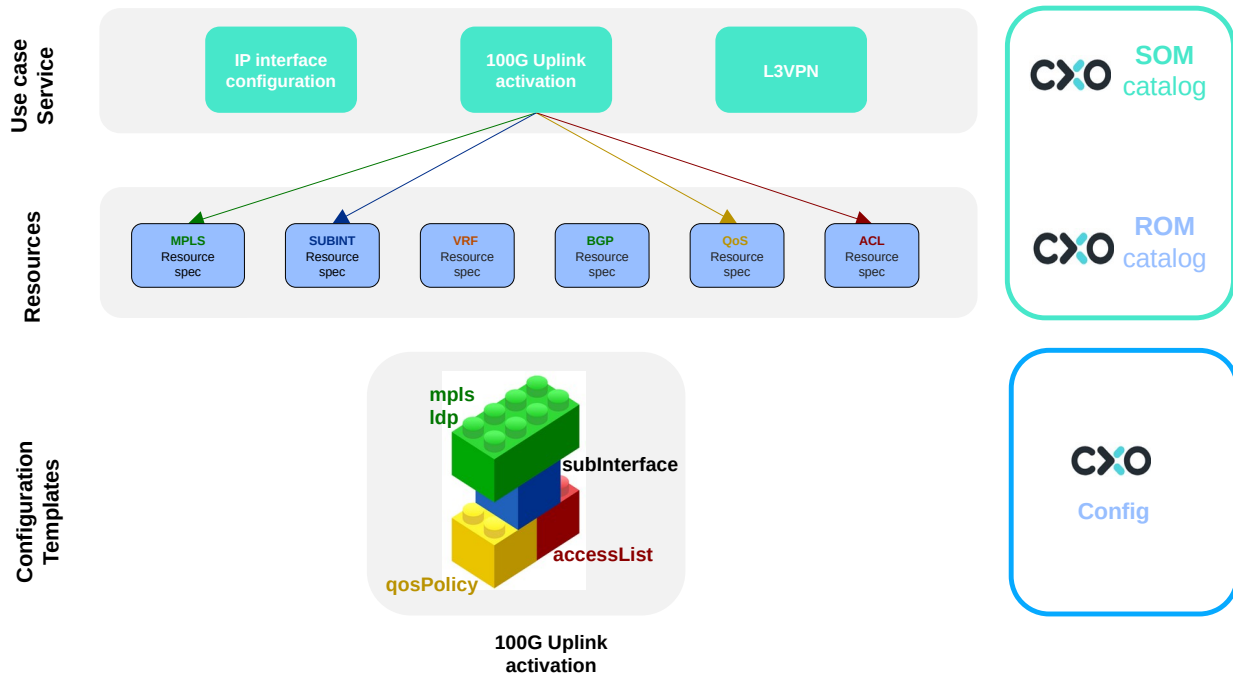
What we actually did

- Observed our network engineers at work
 - Captured the implicit knowledge behind every modeling decision

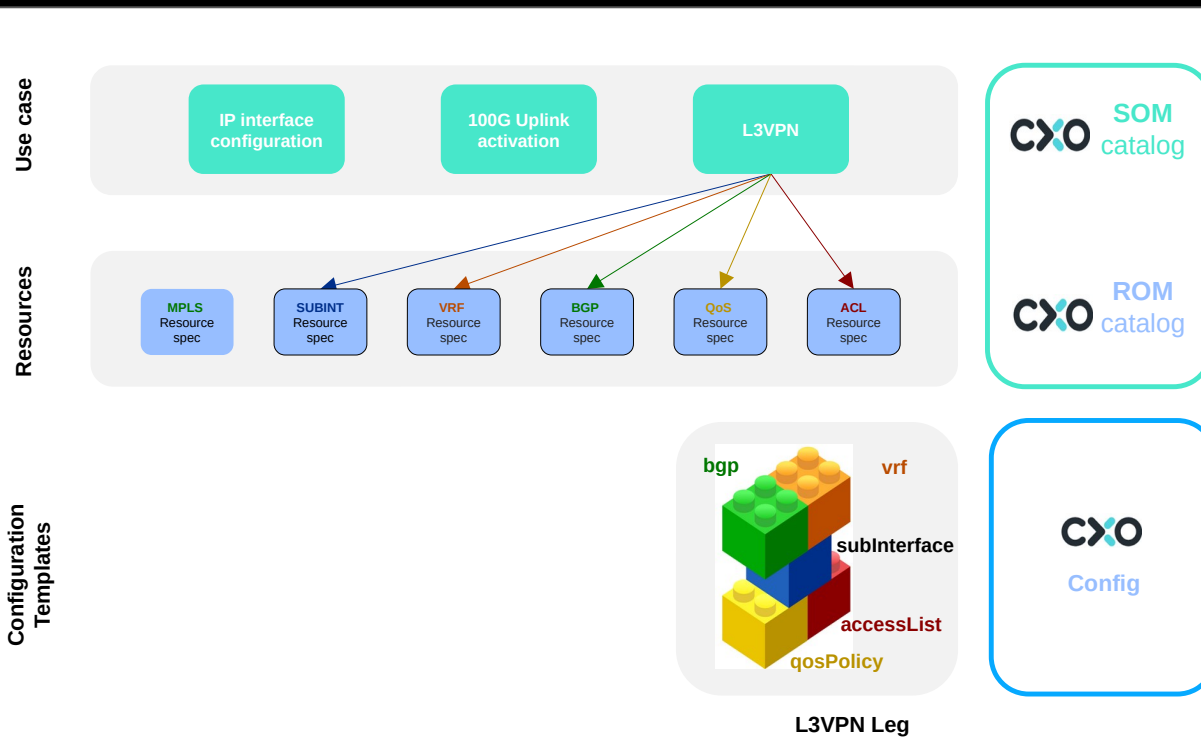
What we actually did

- Identified the design elements that stood the test of time
 - Specifications grounded in schemas
 - Relationships governed by protocol connectivity
 - The **lego concept** of service composition

Lego concept



Lego concept



AI-assisted, expertise-driven
catalog modeling

Conversational modeling

- The network engineer describes the desired catalog in natural language

I want to model link aggregation groups for Nokia SR OS 23.10. Include ports and LACP settings.

Conversational modeling

- The tool explores the schema elements and surfaces those the catalog should be built around

I found the following YANG blocks relevant to your resource. Please confirm which ones you'd like to include:

1. `configure/lag`
2. `configure/lag/lacp`
3. `configure/lag/port`

Please confirm your selection.

12.2 s

Conversational modeling

- The network engineer refines the catalog through conversation

Include port details instead of LACP.

Conversational modeling

- The end catalog supports
 - network discovery
 - provisioning
 - config management workflows

The resource specifications for the Link Aggregation Group (LAG) have been modeled successfully. Here are the details:

Link Aggregation Group (LAG)

- **Description:** This resource specification defines the properties and configuration of a Link Aggregation Group, including its name, encapsulation type, MAC address, operating mode, administrative state, and maximum number of ports.
- **Type:** LogicalResourceSpecification
- **YANG Path:** `configure/lag`
- **Characteristics:**
 - **lag-name** (Type: string)
 - **encap-type** (Type: string)
 - **mac-address** (Type: string)
 - **mode** (Type: string)
 - **admin-state** (Type: string)
 - **max-ports** (Type: integer)

Link Aggregation Group Port

- **Description:** Represents a port within a Link Aggregation Group, including its identifier and priority level.
- **Type:** LogicalResourceSpecification

The bottleneck was always human.

Faster SME onboarding

- TMF concepts and best practices embedded in the process
- Experts contribute meaningfully from day one

Reduced catalog refinements

- Less back and forth between the network engineer and the development team
- The catalog holds up under implementation pressure

Modeling beyond the primary vendor

- The LLM translates concepts and suggests relevant modeling elements across vendors

I'm coming from Nokia SR OS where link aggregation groups are modeled under `configure/lag` — I can set the LAG name, admin-state, and LACP mode there, and assign member ports via `configure/port`. I need to model the equivalent for a Cisco IOS XE 17.12 device. What YANG paths should I be working with?

I found the following YANG blocks relevant to your resource. Please confirm which ones you'd like to include:

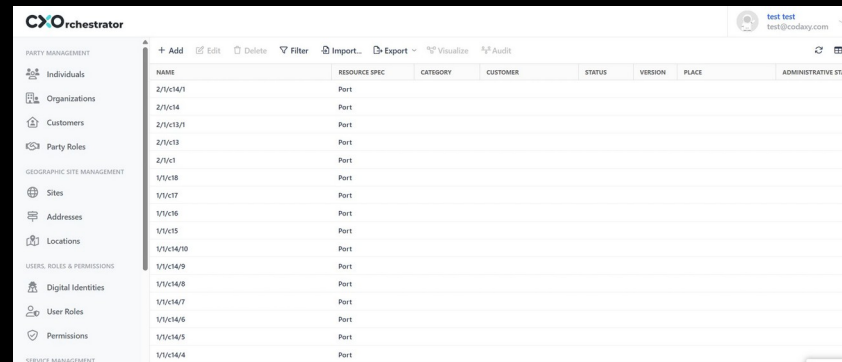
1. `native/interface/Port-channel`
2. `native/interface/GigabitEthernet/channel-group`

Please confirm your selection.

5.2 s

Automated network discovery

- Discovery workflow emerges naturally from the catalog
- Config extraction via NETCONF, CLI scraping, or config dumps becomes trivial



The screenshot shows the CXO Orchestrator web interface. On the left is a sidebar with a navigation menu containing icons and labels for: Individuals, Organizations, Customers, Party Roles, Geographic Site Management (Sites, Addresses, Locations), Users, Roles & Permissions (Digital Identities, User Roles, Permissions), and Service Management. The main area displays a table with columns: NAME, RESOURCE SPEC, CATEGORY, CUSTOMER, STATUS, VERSION, PLACE, and ADMINISTRATIVE STATE. The table contains 15 rows of data, all with 'Port' in the RESOURCE SPEC column. The top of the interface includes a header with the CXO Orchestrator logo, a user profile 'test test' with email 'test@codacy.com', and a toolbar with buttons for '+ Add', 'Edit', 'Delete', 'Filter', 'Import', 'Export', 'Visualize', and 'Audit'.

NAME	RESOURCE SPEC	CATEGORY	CUSTOMER	STATUS	VERSION	PLACE	ADMINISTRATIVE STATE
2/V/14/1	Port						
2/V/14	Port						
2/V/13/1	Port						
2/V/13	Port						
2/V/1	Port						
1/V/18	Port						
1/V/17	Port						
1/V/16	Port						
1/V/15	Port						
1/V/14/10	Port						
1/V/14/9	Port						
1/V/14/8	Port						
1/V/14/7	Port						
1/V/14/6	Port						
1/V/14/5	Port						
1/V/14/4	Port						

Instead of automating
an entire business process
end to end...

..automate the scaffolding
around your domain experts.

The bottleneck was always human.
So is the solution.



Thank you!